

WATER BASED CLASSIFICATION

ML based Water Quality Classification System using random forest and
Support Vector Machine

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

A.Y:2026-27

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DEPARTMENT OF ~ COMPUTER SCIENCE AND ENGINEERING

NAME OF THE TITLE ~ WATER BASED CLASSIFICATION

1. ABSTRACT:

Water is one of the most valuable natural resources, and access to clean drinking water is essential for maintaining public health and ensuring sustainable development. Rapid industrialization, urbanization, and environmental pollution have significantly affected water quality, making continuous monitoring a global necessity. Conventional laboratory-based water quality analysis provides reliable results but often requires specialized equipment, trained personnel, and considerable time, making it unsuitable for quick preliminary assessments. This project proposes an intelligent Machine Learning-based Water Potability Prediction and Classification System capable of determining whether a water sample is potable or non-potable using physicochemical characteristics. The project utilizes the **Water Potability Dataset** from **Kaggle**, which consists of water quality parameters including pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, Turbidity, and the Potability label. Data preprocessing techniques such as handling missing values, feature analysis, and data normalization will be applied before model development. Two supervised machine learning algorithms, **Random Forest** and **Support Vector Machine (SVM)**, will be implemented and their performances compared. The effectiveness of both models will be measured using **Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC Curve, and ROC-AUC Score**. The best-performing model will be integrated into a user-friendly application that predicts water potability based on user-provided inputs. The proposed system demonstrates how machine learning can support faster and cost-effective preliminary water quality assessment while providing practical exposure to the complete machine learning lifecycle, including data preprocessing, model training, evaluation, and deployment.

2. INTRODUCTION:

Access to safe drinking water remains one of the most significant challenges worldwide, as millions of people continue to face health risks caused by contaminated water sources. Water pollution resulting from industrial discharge, agricultural chemicals, untreated sewage, and environmental degradation has increased the demand for efficient and intelligent water quality monitoring systems. Although laboratory testing provides highly accurate results, it is often expensive, time-consuming, and inaccessible in many situations where rapid preliminary assessment is required. These challenges motivate the development of an automated system capable of assisting users in identifying potential water quality conditions using modern machine learning techniques.

Machine Learning has transformed the way prediction and classification problems are solved by enabling computers to identify hidden patterns from historical data instead of relying solely on manually defined rules. In this project, the Water Potability Dataset is used to train Random Forest and Support Vector Machine models for classifying water samples as potable or non-potable based on their physicochemical properties. The project focuses on developing an accurate predictive model, comparing multiple classification algorithms, and integrating the best-performing model into an interactive application that delivers instant predictions. Besides demonstrating the practical application of machine learning in environmental monitoring, the project also provides a

comprehensive understanding of data preprocessing, feature engineering, supervised learning, model evaluation, and real-world deployment.

3. SOFTWARE REQUIREMENTS:

The proposed system requires a software environment capable of performing data preprocessing, machine learning model development, evaluation, visualization, and deployment. Python has been selected as the primary programming language due to its extensive ecosystem of scientific computing and machine learning libraries. The project will be developed using Visual Studio Code or Jupyter Notebook for experimentation and model development, while Flask will be used to deploy the trained model as a lightweight web application. A standard personal computer with moderate hardware specifications is sufficient to execute the project efficiently and support model training, testing, and deployment.

CATEGORY	SPECIFICATION
Operating System	Windows 10/11 (64-bit)
Processor	Intel Core i5 (10th Gen or above) / AMD Ryzen 5 or above
RAM	Minimum 8 GB (16 GB Recommended)
Storage	20 GB Free Disk Space
Programming Language	Python 3.11 or above
Development Environment	Visual Studio Code / Jupyter Notebook
Machine Learning Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Dataset	Water Potability Dataset (Kaggle)
Machine Learning Algorithms	Random Forest, Support Vector Machine (SVM)
Backend Framework	Flask
Frontend Technologies	HTML5, CSS3, JavaScript
Version Control	Git & GitHub

Signature of the guide:

Course Instructor: